

LilyPond

The music typesetter

Extending

The LilyPond development team

This file explains how to extend the functionality of LilyPond version 2.27.2.

For more information about how this manual fits with the other documentation, or to read this manual in other formats, see *Manuals in General Information*.
If you are missing any manuals, the complete documentation can be found at <https://lilypond.org/>.

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For LilyPond version 2.27.2

LilyPond — Extending

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1 Scheme tutorial

LilyPond uses the Scheme programming language, both as part of the input syntax and as the internal mechanism to glue modules of the program together. This section is a very brief overview of entering data in Scheme. If you want to know more about Scheme in general, see <https://www.schemers.org>.

LilyPond uses the GNU Guile implementation of Scheme, which is based on the Scheme “R5RS” standard. If you are learning Scheme to use with LilyPond, working with a different implementation (or referring to a different standard) is not recommended. The Guile Reference Manual can be found at https://www.gnu.org/software/guile/manual/html_node/index.html; this should be the first reference to look up. The “R5RS” Scheme standard itself is located at <https://www.schemers.org/Documents/Standards/R5RS/>.

1.1 Introduction to Scheme

We begin with an introduction to Scheme. For this brief introduction, we will use the Guile interpreter to explore how the language works. Once we are familiar with Scheme, we will show how the language can be integrated in LilyPond files.

1.1.1 Scheme sandbox

The LilyPond installation includes the Guile implementation of Scheme. A hands-on Scheme sandbox with all of LilyPond loaded is available with this call on the command line:

```
lilypond scheme-sandbox
```

Once the sandbox is running, you get the Guile prompt

```
lily-guile@(>)
```

to enter Scheme expressions. In the following, however, we use the shorter string `guile>` to indicate the prompt.

The file containing the sandbox code (`ly/scheme-sandbox.ly`) tries to activate the GNU readline library for nicer editing on the prompt (for example, using the up- and down-arrow keys to access previously typed commands). Currently, this only works with a LilyPond binary dynamically linked with Guile, which is not the case for the ‘officially’ distributed binaries. Two possible workarounds are as follows.

.

Use `rlwrap`, a thin wrapper around the readline library. This program is available for Unix-like operating systems; call it with

```
rlwrap lilypond scheme-sandbox
```

•

Use the Emacs editor in combination with the [LilyPond Tree-sitter Mode](#). This works under Windows, too.

1.1.2 Scheme variables

Scheme variables can have any valid Scheme value, including a Scheme procedure.

Scheme variables are created with `define`:

```
guile> (define a 2)
guile>
```

Scheme variables can be evaluated at the guile prompt simply by typing the variable name:

```
guile> a
$1 = 2
guile>
```

The dollar sign followed by a number is a reference to the result of the just evaluated expression (the variable ‘a’), which you can find on the right side of the equal sign; it can be used later on for evaluation:

```
guile> (+ $1 $1)
$2 = 4
guile>
```

If a Scheme expression evaluates to nothing (as does the call to `define` above), no such reference is shown. Don’t be worried if the number after the dollar sign shown in this tutorial differs from what you get while playing around with the sandbox.

Scheme variables can be printed on the display by using the `display` function:

```
guile> (display a)
2guile>
```

Note that both the value 2 and the guile prompt `guile` showed up on the same line. This can be avoided by calling the `newline` procedure or displaying a newline character.

```
guile> (display a)(newline)
2
guile> (display a)(display "\n")
2
guile>
```

Once a variable has been created, its value can be changed with `set!`:

```
guile> (set! a 12345)
```

```
guile> a
$3 = 12345
guile>
```

1.1.3 Scheme simple data types

The most basic concept in a language is data typing: numbers, character strings, lists, etc. Here is a list of simple Scheme data types that are often used with LilyPond.

Booleans

Boolean values are True or False. The Scheme for True is `#t` and False is `#f`.

Numbers

Numbers are entered in the standard fashion, 1 is the (integer) number one, while -1.5 is a floating point number (a non-integer number).

Strings

Strings are enclosed in double quotes:

```
"this is a string"
```

Strings may span several lines:

```
"this
is
a string"
```

and the newline characters at the end of each line will be included in the string.

Newline characters can also be added by including `\n` in the string.

```
"this\nis a\nmultiline string"
```

Quotation marks and backslashes are added to strings by preceding them with a backslash. The string `\a said "b"` is entered as

```
"\\a said \\\"b\\\""
```

There are additional Scheme data types that are not discussed here. For a complete listing see the Guile reference guide,

https://www.gnu.org/software/guile/manual/html_node/Data-Types.html.

1.1.4 Scheme compound data types

There are also compound data types in Scheme. The types commonly used in LilyPond programming include pairs, lists, alists, and hash tables.

Pairs

The foundational compound data type of Scheme is the `pair`. As might be expected from its name, a pair is two values glued together. The operator used to form a pair is called `cons`.

```
guile> (cons 4 5)
$4 = (4 . 5)
guile>
```

Note that the pair is displayed as two items surrounded by parentheses and separated by whitespace, a period (.), and more whitespace. The period is *not* a decimal point, but rather an indicator of the pair.

Pairs can also be entered as literal values by preceding them with a single quote character.

```
guile> '(4 . 5)
$5 = (4 . 5)
guile>
```

The two elements of a pair may be any valid Scheme value:

```
guile> (cons #t #f)
$6 = (#t . #f)
guile> '("blah-blah" . 3.1415926535)
$7 = ("blah-blah" . 3.1415926535)
guile>
```

The first and second elements of the pair can be accessed by the Scheme procedures `car` and `cdr`, respectively. The dots shown instead of the Guile prompt indicate that the just entered expression is not complete yet.

```
guile> (define mypair (cons 123 "hello there"))
... )
guile> (car mypair)
$8 = 123
guile> (cdr mypair)
$9 = "hello there"
guile>
```

Note: `cdr` is pronounced "could-er", according Sussman and Abelson, see https://mitpress.mit.edu/sites/default/files/sicp/full-text/book/book-Z-H-14.html#footnote_Temp_133

Lists

A very common Scheme data structure is the *list*. Formally, a 'proper' list is defined to be either the empty list with its input form `()` and length 0, or a pair whose `cdr` in turn is a shorter list.

There are many ways of creating lists. Perhaps the most common is with the `list` procedure:

```
guile> (list 1 2 3 "abc" 17.5)
$10 = (1 2 3 "abc" 17.5)
guile>
```


Representing a list as individual elements separated by whitespace and enclosed in parentheses is actually a compacted rendition of the actual dotted pairs constituting the list, where the dot and an immediately following starting paren are removed along with the matching closing paren. Without this compaction, the output would have been

```
(1 . (2 . (3 . ("abc" . (17.5 . ())))))
```

As with the output, a list can also be entered (after adding a quote to avoid interpretation as a function call) as a literal list by enclosing its elements in parentheses:

```
guile> '(17 23 "foo" "bar" "bazzle")
$11 = (17 23 "foo" "bar" "bazzle")
guile>
```

Lists are a central part of Scheme. In, fact, Scheme is considered a dialect of lisp, where ‘lisp’ is an abbreviation for ‘List Processing’. Scheme expressions are all lists.

Association lists (alists)

A special type of list is an *association list* or *alist*. An alist is used to store data for easy retrieval.

Alists are lists whose elements are pairs. The *car* of each element is called the *key*, and the *cdr* of each element is called the *value*. The Scheme procedure `assoc` is used to retrieve an entry from the alist, and `cdr` is used to retrieve the value:

```
guile> (define my-alist '((1 . "A") (2 . "B") (3 . "C")))
guile> my-alist
$12 = ((1 . "A") (2 . "B") (3 . "C"))
guile> (assoc 2 my-alist)
$13 = (2 . "B")
guile> (cdr (assoc 2 my-alist))
$14 = "B"
guile>
```

Alists are widely used in LilyPond to store properties and other data.

Hash tables

A data structure that is used occasionally in LilyPond. A hash table is similar to an array, but the indexes to the array can be any type of Scheme value, not just integers.

Hash tables are more efficient than alists if there is a lot of data to store and the data changes very infrequently.

The syntax to create hash tables is a bit complex, but you can see examples of it in the LilyPond source.

```
guile> (define h (make-hash-table 10))
guile> h
$15 = #<hash-table 1b9e0640 0/31>
guile> (hashq-set! h 'key1 "val1")
$16 = "val1"
```

```
guile> (hashq-set! h 'key2 "val2")
$17 "val2"
guile> (hashq-set! h 3 "val3")
$18 = "val3"
guile>
```

(The hexadecimal number 1b9e0640, uniquely identifying the hash, might be different on your computer. We don't need this information here.)

Values are retrieved from hash tables with `hashq-ref`.

```
guile> (hashq-ref h 3)
$19 = "val3"
guile> (hashq-ref h 'key2)
$20 = "val2"
guile>
```

Keys and values are retrieved as a pair with `hashq-get-handle`. This is a preferred way, because it will return `#f` if a key is not found.

```
guile> (hashq-get-handle h 'key1)
$21 = (key1 . "val1")
guile> (hashq-get-handle h 'frob)
$22 = #f
guile>
```

1.1.5 Calculations in Scheme

Scheme can be used to do calculations. It uses *prefix* syntax. Adding 1 and 2 is written as `(+ 1 2)` rather than the traditional `1+2`.

```
guile> (+ 1 2)
$23 = 3
guile>
```

Calculations may be nested; the result of a function may be used for another calculation.

```
guile> (+ 1 (* 3 4))
$24 = 13
guile>
```

These calculations are examples of evaluations; an expression like `(* 3 4)` is replaced by its value 12.

Scheme calculations are sensitive to the differences between integers and non-integers. Integer calculations are exact, while non-integers are calculated to the appropriate limits of precision:

```
guile> (/ 7 3)
$25 = 7/3
guile> (/ 7.0 3.0)
$26 = 2.3333333333333335
guile>
```

When the Scheme interpreter encounters an expression that is a list, the first element of the list is treated as a procedure to be evaluated with the arguments of the remainder of the list. Therefore, all operators in Scheme are prefix operators.

If the first element of a Scheme expression that is a list passed to the interpreter is *not* an operator or procedure, an error will occur:

```
guile> (1 2 3)
ice-9/boot-9.scm:1676:22: In procedure raise-exception:
Wrong type to apply: 1

Entering a new prompt. Type `,bt' for a backtrace or `,q' to continue.
guile [1]> ,q
guile>
```

Here you can see that the interpreter was trying to treat 1 as an operator or procedure, and it couldn't. Hence the error is "Wrong type to apply: 1".

Therefore, to create a list we need to use the list operator, or to quote the list so that the interpreter will not try to evaluate it.

```
guile> (list 1 2 3)
$27 = (1 2 3)
guile> '(1 2 3)
$28 = (1 2 3)
guile>
```

This is an error that can appear as you are working with Scheme in LilyPond.

1.1.6 Scheme procedures

Scheme procedures are executable Scheme expressions that return a value resulting from their execution. They can also manipulate variables defined outside of the procedure.

Defining procedures

Procedures are defined in Scheme with `define`

```
(define (function-name arg1 arg2 ... argn)
  scheme-expression-that-gives-a-return-value)
```

For example, we could define a procedure to calculate the average:

```
guile> (define (average x y) (/ (+ x y) 2))
guile> average
$29 = #<procedure average (x y)>
guile>
```

Once a procedure is defined, it is called by putting the procedure name and the arguments in a list. For example, we can calculate the average of 3 and 12:

```
guile> (average 3 12)
$30 = 15/2
```

```
guile>
```

Predicates

Scheme procedures that return boolean values are often called *predicates*. By convention (but not necessity), predicate names typically end in a question mark:

```
guile> (define (less-than-ten? x) (< x 10))
guile> (less-than-ten? 9)
$31 = #t
guile> (less-than-ten? 15)
$32 = #f
guile>
```

Return values

Scheme procedures always return a return value, which is the value of the last expression executed in the procedure. The return value can be any valid Scheme value, including a complex data structure or a procedure.

Sometimes the user would like to have multiple Scheme expressions in a procedure. There are two ways that multiple expressions can be combined. The first is the `begin` procedure, which allows multiple expressions to be evaluated, and returns the value of the last expression.

```
guile> (begin (+ 1 2) (- 5 8) (* 2 2))
$33 = 4
guile>
```

The second way to combine multiple expressions is in a `let` block. In a `let` block, a series of bindings are created, and then a sequence of expressions that can include those bindings is evaluated. The return value of the `let` block is the return value of the last statement in the `let` block:

```
guile> (let ((x 2) (y 3) (z 4)) (display (+ x y)) (display (- z 4))
... (+ (* x y) (/ z x)))
50$34 = 8
guile>
```

1.1.7 Scheme conditionals

if

Scheme has an `if` procedure:

```
(if test-expression true-expression false-expression)
```

test-expression is an expression that returns a boolean value. If *test-expression* returns `#t`, the `if` procedure returns the value of *true-expression*, otherwise it returns the value of *false-expression*.

```
guile> (define a 3)
```

```
guile> (define b 5)
guile> (if (> a b) "a is greater than b" "a is not greater than b")
$35 = "a is not greater than b"
guile>
```

cond

Another conditional procedure in Scheme is `cond`:

```
(cond (test-expression-1 result-expression-sequence-1)
      (test-expression-2 result-expression-sequence-2)
      ...
      (test-expression-n result-expression-sequence-n))
```

For example:

```
guile> (define a 6)
guile> (define b 8)
guile> (cond ((< a b) "a is less than b")
... ((= a b) "a equals b")
... ((> a b) "a is greater than b"))
$36 = "a is less than b"
guile>
```

1.2 Scheme in LilyPond

1.2.1 LilyPond Scheme syntax

The Guile interpreter is part of LilyPond, which means that Scheme can be included in LilyPond input files. There are several methods for including Scheme in LilyPond.

The simplest way is to use a hash mark `#` before a Scheme expression.

Now LilyPond's input is structured into tokens and expressions, much like human language is structured into words and sentences. LilyPond has a lexer that recognizes tokens (literal numbers, strings, Scheme elements, pitches and so on), and a parser that understands the syntax. Once it knows that a particular syntax rule applies, it executes actions associated with it.

The hash mark `#` method of embedding Scheme is a natural fit for this system. Once the lexer sees a hash mark, it calls the Scheme reader to read one full Scheme expression (this can be an identifier, an expression enclosed in parentheses, or several other things). After the Scheme expression is read, it is stored away as the value for an `SCM_TOKEN` in the grammar. Once the parser knows how to make use of this token, it calls Guile for evaluating the Scheme expression. Since the parser usually requires a bit of lookahead from the lexer to make its parsing decisions, this separation of reading and evaluation between lexer and parser is exactly what is needed to keep the execution of LilyPond and Scheme expressions in sync. For this reason, you should use the hash mark `#` for calling Scheme whenever this is feasible.

Another way to call the Scheme interpreter from LilyPond is the use of dollar \$ instead of a hash mark for introducing Scheme expressions. In this case, LilyPond evaluates the code right after the lexer has read it. It checks the resulting type of the Scheme expression and then picks a token type (one of several `xxx_IDENTIFIER` in the syntax) for it. It creates a copy of the value and uses that for the value of the token. If the value of the expression is void (Guile's value of `*unspecified*`), nothing at all is passed to the parser.

This is, in fact, exactly the same mechanism that LilyPond employs when you call any variable or music function by name, as `\name`, with the only difference that the name is determined by the LilyPond lexer without consulting the Scheme reader, and thus only variable names consistent with the current LilyPond mode are accepted.

The immediate action of \$ can lead to surprises, see [Importing Scheme in LilyPond](#). Using # where the parser supports it is usually preferable. Inside of music expressions, expressions created using # *are* interpreted as music. However, they are *not* copied before use. If they are part of some structure that might still get used, you may need to use `ly:music-deep-copy` explicitly.

There are also 'list splicing' operators \$@ and #@ that insert all elements of a list in the surrounding context.

Now let's take a look at some actual Scheme code. Scheme procedures can be defined in LilyPond input files:

```
#(define (average a b c) (/ (+ a b c) 3))
```

Note that LilyPond comments (% and %{ %}) cannot be used within Scheme code, even in a LilyPond input file, because the Guile interpreter, not the LilyPond lexer, is reading the Scheme expression. Comments in Guile Scheme are entered as follows:

```
; this is a single-line comment

#!
  This a (non-nestable) Guile-style block comment
  But these are rarely used by Schemers and never in
  LilyPond source code
!#
```

For the rest of this section, we will assume that the data is entered in a music file, so we add a # at the beginning of each Scheme expression.

All of the top-level Scheme expressions in a LilyPond input file can be combined into a single Scheme expression by use of the `begin` statement:

```
#(begin
  (define foo 0)
  (define bar 1))
```

1.2.2 LilyPond variables

LilyPond variables are stored internally in the form of Scheme variables. Thus,

```
twelve = 12
```

is equivalent to

```
 #(define twelve 12)
```

This means that LilyPond variables are available for use in Scheme expressions. For example, we could use

```
twentyFour = #(* 2 twelve)
```

which would result in the number 24 being stored in the LilyPond (and Scheme) variable `twentyFour`.

Scheme allows modifying complex expressions in-place and LilyPond makes use of this ‘in-place modification’ when using music functions. But when music expressions are stored in variables rather than entered directly the usual expectation, when passing them to music functions, would be that the original value is unmodified. So when referencing a music variable with leading backslash (such as `\twentyFour`), LilyPond creates a copy of that variable’s music value for use in the surrounding music expression rather than using the variable’s value directly.

Therefore, Scheme music expressions written with the `#` syntax should be used for material that is created ‘from scratch’ (or that is explicitly copied) rather than being used, instead, to directly reference material.

See also

Extending: [LilyPond Scheme syntax](#).

1.2.3 Debugging Scheme code

When debugging large Scheme programs, it is helpful when an error occurs to be pointed to the precise line in the program where it was raised. In order to enable source locations to be shown for errors in Scheme code, LilyPond must be run with the command-line option `-dcompile-scheme-code`. Alternatively, writing `#(ly:set-option 'compile-scheme-code)` at the top of the LilyPond file has the same effect.

Furthermore, to obtain even more information about the error, you may use the `-ddebug-eval` option, or the code `#(debug-enable 'backtrace)` in the file. Under this mode, when an error occurs, a verbose “backtrace” is displayed, containing information about all function calls that led to the problematic spot.

Known issues and warnings

Due to a limitation in the Guile implementation, the `-dcompile-scheme-code` option currently makes it impossible to compile LilyPond files with more than a few thousand Scheme expressions.

1.2.4 Input variables and Scheme

The input format supports the notion of variables: in the following example, a music expression is assigned to a variable with the name `traLaLa`.

```
traLaLa = { c'4 d'4 }
```

There is also a form of scoping: in the following example, the `\layout` block also contains a `traLaLa` variable, which is independent of the outer `\traLaLa`.

```
traLaLa = { c'4 d'4 }
\layout { traLaLa = 1.0 }
```

In effect, each input file is a scope, and all `\header`, `\midi`, and `\layout` blocks are scopes nested inside that toplevel scope.

Both variables and scoping are implemented in the Guile module system. An anonymous Scheme module is attached to each scope. An assignment of the form:

```
traLaLa = { c'4 d'4 }
```

is internally converted to a Scheme definition:

```
(define traLaLa Scheme value of '...')
```

This means that LilyPond variables and Scheme variables may be freely mixed. In the following example, a music fragment is stored in the variable `traLaLa`, and duplicated using Scheme. The result is imported in a `\score` block by means of a second variable `twice`:

```
traLaLa = { c'4 d'4 }

#(define newLa (map ly:music-deep-copy
  (list traLaLa traLaLa)))
#(define twice
  (make-sequential-music newLa))

\twice
```



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This is actually a rather interesting example. The assignment will only take place after the parser has ascertained that nothing akin to `\addlyrics` follows, so it needs to check what

comes next. It reads `#` and the following Scheme expression *without* evaluating it, so it can go ahead with the assignment, and *afterwards* execute the Scheme code without problem.

1.2.5 Importing Scheme in LilyPond

The above example shows how to ‘export’ music expressions from the input to the Scheme interpreter. The opposite is also possible. By placing it after `$`, a Scheme value is interpreted as if it were entered in LilyPond syntax. Instead of defining `\twice`, the example above could also have been written as

```
...
$(make-sequential-music newLa)
```

You can use `$` with a Scheme expression anywhere you could use `\name` after having assigned the Scheme expression to a variable *name*. This replacement happens in the ‘lexer’, so LilyPond is not even aware of the difference.

One drawback, however, is that of timing. If we had been using `$` instead of `#` for defining `newLa` in the above example, the following Scheme definition would have failed because `traLaLa` would not yet have been defined. For an explanation of this timing problem, [LilyPond Scheme syntax](#).

A further convenience can be the ‘list splicing’ operators `$@` and `#@` for inserting the elements of a list in the surrounding context. Using those, the last part of the example could have been written as

```
...
{ #@newLa }
```

Here, every element of the list stored in `newLa` is taken in sequence and inserted into the list, as if we had written

```
{ #(first newLa) #(second newLa) }
```

Now in all of these forms, the Scheme code is evaluated while the input is still being consumed, either in the lexer or in the parser. If you need it to be executed at a later point of time, check out [Void Scheme functions](#), or store it in a procedure:

```
...(define (nopc)
      (ly:set-option 'point-and-click #f))

...
#(nopc)
{ c'4 }
```

1.2.6 Object properties

Object properties are stored in LilyPond in the form of alist-chains, which are lists of alists. Properties are set by adding values at the beginning of the property list. Properties are read by retrieving values from the alists.

Setting a new value for a property requires assigning a value to the alist with both a key and a value. The LilyPond syntax for doing this is:

```
\override Stem.thickness = #2.6
```

This instruction adjusts the appearance of stems. An alist entry '(thickness . 2.6)' is added to the property list of the Stem object. thickness is measured relative to the thickness of staff lines, so these stem lines will be 2.6 times the width of staff lines. This makes stems almost twice as thick as their normal size. To distinguish between variables defined in input files (like twentyFour in the example above) and variables of internal objects, we will call the latter 'properties' and the former 'variables.' So, the stem object has a thickness property, while twentyFour is a variable.

1.2.7 LilyPond compound variables

Offsets

Two-dimensional offsets (X and Y coordinates) are stored as *pairs*. The car of the offset is the X coordinate, and the cdr is the Y coordinate.

```
\override TextScript.extra-offset = #'(1 . 2)
```

This assigns the pair (1 . 2) to the extra-offset property of the TextScript object. These numbers are measured in staff-spaces, so this command moves the object 1 staff space to the right, and 2 spaces up.

Procedures for working with offsets are found in scm/lily-library.scm.

Fractions

Fractions as used by LilyPond are again stored as *pairs*, this time of unsigned integers. While Scheme can represent rational numbers as a native type, musically '2/4' and '1/2' are not the same, and we need to be able to distinguish between them. Similarly there are no negative 'fractions' in LilyPond's mind. So 2/4 in LilyPond means (2 . 4) in Scheme, and #2/4 in LilyPond means 1/2 in Scheme.

Extents

Pairs are also used to store intervals, which represent a range of numbers from the minimum (the car) to the maximum (the cdr). Intervals are used to store the X- and Y-extents of printable objects. For X extents, the car is the left hand X coordinate, and the cdr is the right hand X coordinate. For Y extents, the car is the bottom coordinate, and the cdr is the top coordinate.

Procedures for working with intervals are found in scm/lily-library.scm. These procedures should be used when possible to ensure consistency of code.

Property alists

A property alist is a LilyPond data structure that is an alist whose keys are properties and whose values are Scheme expressions that give the desired value for the property.

LilyPond properties are Scheme symbols, such as `'thickness`.

Alist chains

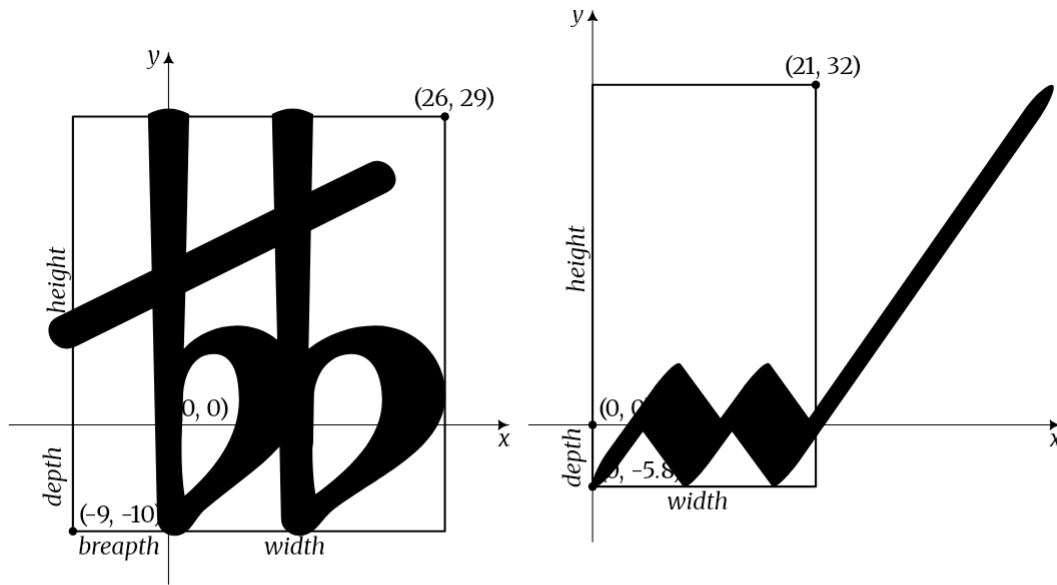
An alist chain is a list containing property alists.

The set of all properties that will apply to a grob is typically stored as an alist chain. In order to find the value for a particular property that a grob should have, each alist in the chain is searched in order, looking for an entry containing the property key. The first alist entry found is returned, and the value is the property value.

The Scheme procedure `chain-assoc-get` is normally used to get grob property values.

1.2.8 LilyPond's box model

The 'classical' box model for text (as used, for example, in TeX), comprises the three dimensions *width*, *height*, and *depth*: glyphs are positioned on a horizontal text base line; the height of a glyph is the vertical size above this line, the depth the size below this line, and the width holds the horizontal size of the glyph. LilyPond adds a fourth dimension, *breapth*,¹ to extend the horizontal size by a part left of the glyph's 'reference point', denoted with '(0, 0)' in the image below. The main reason for this extension is that many glyphs like fermatas are symmetrical horizontally (and also symmetrically attached to a note head), and the best reference point for such glyphs is consequently the horizontal center. Other glyphs like the quarter-tone accidental shown on the left in the image also have considerable *breapth*.



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As demonstrated by the custos glyph on the right side of the image, the box defined for a glyph often differs considerably of what gets actually printed: LilyPond positions the right side of the custos glyph box at the end of a staff, making the long extension line stick out to the right into the margin.

This box model is not only used for LilyPond's musical glyphs but also for markup of any kind. Both the horizontal and vertical dimensions are specified as an extent; using Scheme notation, the left glyph in the image above has a horizontal extent of `#'(-9 . 26)` and a vertical extent of `#'(-10 . 29)`.

LilyPond also uses some extensions to the box model to cover special cases.

-

If a box should not take part in LilyPond's music spacing algorithm along a certain axis, set the extents of this axis to the otherwise nonsensical value `#'(+inf.0 . -inf.0)`, also available as the symbol `empty-interval`.

-

In markup, *spacing* (which normally causes horizontal or vertical movement of the next object in the markup) is also set up as a box internally: the extent along the orthogonal axis is set to `empty-interval` as a marker, and a spacing value *s* is represented as the extent `#' (0 . s)`. Contrary to ‘normal’ boxes, spacing can be also negative: for example, the horizontal extent of the markup command `\hspace #-4` is `#' (0 . -4)`.

1.2.9 Internal music representation

Internally, music is represented as a Scheme list. The list contains various elements that affect the printed output. Parsing is the process of converting music from the LilyPond input representation to the internal Scheme representation.

When a music expression is parsed, it is converted into a set of Scheme music objects. The defining property of a music object is that it takes up time. The time it takes up is called its *duration*. Durations are expressed as a rational number that measures the length of the music object in whole notes.

A music object has three kinds of types:

- music name: Each music expression has a name. For example, a note leads to a `NoteEvent` in *Internals Reference*, and `\simultaneous` leads to a `SimultaneousMusic` in *Internals Reference*. A list of all expressions available is in the *Internals Reference* manual, under Music expressions in *Internals Reference*.
- ‘type’ or interface: Each music name has several ‘types’ or interfaces, for example, a note is an event, but it is also a note-event, a rhythmic-event, and a melodic-event. All classes of music are listed in the *Internals Reference*, under Music classes in *Internals Reference*.
- C++ object: Each music object is represented by an object of the C++ class `Music`.

The actual information of a music expression is stored in properties. For example, a `NoteEvent` in *Internals Reference* has `pitch` and `duration` properties that store the pitch and duration of that note. A list of all properties available can be found in the *Internals Reference*, under Music properties in *Internals Reference*.

A compound music expression is a music object that contains other music objects in its properties. A list of objects can be stored in the `elements` property of a music object, or a single ‘child’ music object in the `element` property. For example, `SequentialMusic` in *Internals Reference* has its children in `elements`, and `GraceMusic` in *Internals Reference* has its single argument in `element`. The body of a repeat is stored in the `element` property of `VoltaRepeatedMusic` in *Internals Reference*, and the alternatives in `elements`.

1.3 Building complicated functions

This section explains how to gather the information necessary to create complicated music functions.

1.3.1 A first approach (example)

1.

The problem

Using the following input, the ‘espressivo’ sign (i.e., the two small hairpins) should become wider, but not taller.

```
{ f' \espressivo }
```



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2.

Possible solutions

A.

create a new stencil

B.

scale the default stencil in direction of the X-axis

3.

Thoughts about previous item

Solution A is doable. However, it will result in a higher coding effort. On the other hand, solution B is cheaper, but it has the disadvantage that it doesn't print acceptable results in all cases (although you may not know this before you've actually tried).

☒ choose solution B.

4.

Coding

o

If you want to manipulate the default stencil you have to find information about it.

The `espressivo` sign is a `Script grob`. Looking this up in the IR (*Script in Internals Reference*), we see

```
'stencil' (stencil):
  'ly:script-interface::print'
```

Let us test whether this is correct. To be sure that we affect the correct grob we are going to use colors. There is a predefined function for that, see Scheme functions in *Internals Reference*.

```
-- Function: stencil-with-color stencil color
Return a modified version of the given stencil that is
colored with the given color. See 'normalize-color' for
possible color formats.
```

```
-- Function: normalize-color color
Convert a color given in any of the supported formats into
a list of 4 numbers: R, G, B, A. Possible formats are:
such a list of 4 numbers; a list of 3 numbers (transparency
defaults to 1.0); a CSS string (named color, or "#RRGGBB",
or "#RRGGBBAA", or "#RGB", or "#RGBA").
```

Using a named color ('red'), this results in the following code.

```
colorDefaultStencil =
\once \override Script.stencil =
% `lambda` starts a procedure; its argument is `grob`.
#(lambda (grob)
  ;; With `let`, local variables are defined.
  (let ((stil (ly:script-interface::print grob)))
    ;; The procedure returns the colored default stencil.
    (stencil-with-color stil red)))

{ \colorDefaultStencil f'\espressivo }
```



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Fine, works.

o

This kind of operation – taking a default stencil or property and returning a modified version of it – is very common, and LilyPond provides a special function for it: `grob-transformer`.

```
-- Function: grob-transformer property func
Create an override value good for applying FUNC to either
pure or unpure values. FUNC is called with the
respective grob as first argument and the default value
(after resolving all callbacks) as the second.
```

Let's try that. (The concept of pure vs. unpure gets explained later, see [Unpure-pure containers](#); we can ignore it for now).

```
colorDefaultStencil =
\once \override Script.stencil =
  #(grob-transformer 'stencil
    (lambda (grob orig)
      (stencil-with-color orig red)))

{ \colorDefaultStencil f'\espressivo }
```



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Success!

○

To get a scaled stencil we have to use

```
-- Function: ly:stencil-scale stil x y
Scale stencil STIL using the horizontal and vertical
scaling factors X and optional Y (defaulting to X).
Negative values flip or mirror STIL without changing its
origin; this may result in collisions unless it is
repositioned.
```

from the same chapter in the Internals Reference.

Result:

```
scaleColorDefaultStencil =
\once \override Script.stencil =
  #(grob-transformer 'stencil
    (lambda (grob orig)
      (stencil-with-color
        (ly:stencil-scale
          orig
          ;; 1 is the neutral element with ly:stencil-scale,
          ;; i.e., scaling with '1 1' (for x- and y-axis)
          ;; returns a visibly unchanged stencil.
          2 1)
        red)))

{ \scaleColorDefaultStencil f'\espressivo }
```



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Apart from the color that seems to be the desired output.

○

There are some problems, though, as can be seen in the following example


```
{ \scaleColorDefaultStencil f'_\espressivo ^\fermata }
```



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■

The fermata is scaled, too.

■

Having to type the command before the note(s) and `\espressivo` after the note(s) is annoying.

Solution for both: use `\tweak`!

```
{
  f' - \tweak stencil
    #(grob-transformer 'stencil
      (lambda (grob orig)
        (stencil-with-color
          (ly:stencil-scale orig 2 1)
          red)))
    _\espressivo
    ^\fermata
}
```



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This does what we want, however, typing so much isn't nice – let's define a variable and use it with the tweak. With some copy and paste we get

```

#(define longer-script
  (grob-transformer 'stencil
    (lambda (grob orig)
      (stencil-with-color
        (ly:stencil-scale orig 2 1)
        red))))

{
  f' - \tweak stencil #longer-script _\espressivo
  ^\fermata
}
```



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Works fine, though the scaling values are hard-coded, and typing `\tweak stencil #longer-script` repeatedly is still awkward.

To fix that, let's first introduce some variables in `longer-script` for `x` and `y` scaling values.

```
#(define (longer-script x y)
  (grob-transformer 'stencil
    (lambda (grob orig)
      (stencil-with-color
        (ly:stencil-scale orig x y)
        red))))
```

Second, to reduce typing, we define an event function, see [Event functions](#). The elements in the first parenthesis group (`(x-val y-val)`) are the function variables, the elements in the second parenthesis group the respective variable types (two times we check for a number; see *Predefined type predicates* in *Notation Reference* for possible tests).

```
scaleEspr =
#(define-event-function (x-val y-val) (number? number?)
  #{
    \tweak stencil #(longer-script x-val y-val)
    \espressivo
  #})

{ f' _\scaleEspr 2 1
  ^\fermata }
```



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Works as expected, fine! Note the use of `#{ ... #}`: Within this Scheme construction, we can use LilyPond syntax, and the function's arguments (like `x-val`) are properly expanded. See [LilyPond code blocks](#).

5.

The finished code

Finally, we want to do some clean-up.

○

We only want scaling in the `x`-direction, so let's hard-code `y`-scaling in the event function.

○

Delete the color.

This eventually leads to the following code.

```
#(define (longer-script x y)
  (grob-transformer 'stencil
    (lambda (grob orig)
      (ly:stencil-scale orig x y))))

longEspressivo =
#(define-event-function (x-val) (number?)
  #{
    \tweak stencil #(longer-script x-val 1)
    \espressivo
  })

{ f'^\longEspressivo #2 _\fermata }
```



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Voilà!

The main benefit of using `grob-transformer` is that we no longer have to take care of the actual name of the stencil function. In the next example we rotate the stencil of a slur and a beam by 90 degrees – that the actual stencil functions are called `ly:slur::print` and `ly:beam::print`, respectively, is of no concern here.

```
#(define rotate-90
  (grob-transformer 'stencil
    (lambda (grob orig)
      (ly:stencil-rotate orig 90 0 0))))

{ f'8( g')
  f'8-\tweak stencil #rotate-90 ( g')
  f'8[ g']
  f'8-\tweak stencil #rotate-90 [ g'] }
```



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1.3.2 Displaying music expressions

When writing a music function it is often instructive to inspect how a music expression is stored internally. This can be done with the music function `\displayMusic`.

```
{
  \displayMusic { c'4\f }
}
```

will display

```
(make-music
 'SequentialMusic
 'elements
 (list (make-music
        'NoteEvent
        'articulations
        (list (make-music
                'AbsoluteDynamicEvent
                'text
                "f"))
        'duration
        (ly:make-duration 2 0 1/1)
        'pitch
        (ly:make-pitch 0 0 0))))
```

By default, LilyPond will print these messages to the console along with all the other messages. To split up these messages and save the results of `\display{STUFF}`, you can specify an optional output port to use:

```
{
  \displayMusic #(open-output-file "display.txt") { c'4\f }
}
```

This will overwrite a previous output file whenever it is called; if you need to write more than one expression, you would use a variable for your port and reuse it:

```
{
  port = #(open-output-file "display.txt")
  \displayMusic \port { c'4\f }
  \displayMusic \port { d'4 }
  #(close-output-port port)
}
```

Guile's manual describes ports in detail. Closing the port is actually only necessary if you need to read the file before LilyPond finishes; in the first example, we did not bother to do so.

A bit of reformatting makes the above information easier to read:

```
(make-music 'SequentialMusic
 'elements (list
  (make-music 'NoteEvent
    'articulations (list
      (make-music 'AbsoluteDynamicEvent
        'text
        "f"))
    'duration (ly:make-duration 2 0 1/1)
    'pitch (ly:make-pitch 0 0 0))))
```

A `{ ... }` music sequence has the name `SequentialMusic`, and its inner expressions are stored as a list in its `'elements` property. A note is represented as a `NoteEvent` object

(storing the duration and pitch properties) with attached information (in this case, an `AbsoluteDynamicEvent` with a "f" text property) stored in its `articulations` property.

`\displayMusic` returns the music it displays, so it will get interpreted as well as displayed. To avoid interpretation, write `\void` before `\displayMusic`.

1.3.3 Music properties

Let's look at an example:

```
someNote = c'
\displayMusic \someNote
====>
(make-music
 'NoteEvent
 'duration
 (ly:make-duration 2 0 1/1)
 'pitch
 (ly:make-pitch 0 0 0))
```

The `NoteEvent` object is the representation of `someNote`. Straightforward. How about putting `c'` in a chord?

```
someNote = <c'>
\displayMusic \someNote
====>
(make-music
 'EventChord
 'elements
 (list (make-music
        'NoteEvent
        'duration
        (ly:make-duration 2 0 1/1)
        'pitch
        (ly:make-pitch 0 0 0))))
```

Now the `NoteEvent` object is the first object of the `'elements` property of `someNote`.

The `display-scheme-music` function is the function used by `\displayMusic` to display the Scheme representation of a music expression.

```
 #(display-scheme-music (first (ly:music-property someNote 'elements)))
====>
(make-music
 'NoteEvent
 'duration
 (ly:make-duration 2 0 1/1)
 'pitch
 (ly:make-pitch 0 0 0))
```

Then the note pitch is accessed through the `'pitch` property of the `NoteEvent` object.

```
 #(display-scheme-music
  (ly:music-property (first (ly:music-property someNote 'elements))
```

```

'pitch))
====>
(ly:make-pitch 0 0 0)

```

The note pitch can be changed by setting this 'pitch property.

```

#(set! (ly:music-property (first (ly:music-property someNote 'elements))
                           'pitch)
      (ly:make-pitch 0 1 0)) ;; set the pitch to d'.
\displayLilyMusic \someNote
====>
d'4

```

1.3.4 Doubling a note with slurs (example)

Suppose we want to create a function that translates input like `a` into `{ a( a) }`. We begin by examining the internal representation of the desired result.

```

\displayMusic{ a'( a') }
====>
(make-music
 'SequentialMusic
 'elements
 (list (make-music
        'NoteEvent
        'articulations
        (list (make-music
                'SlurEvent
                'span-direction
                -1))
        'duration
        (ly:make-duration 2 0 1/1)
        'pitch
        (ly:make-pitch 0 5 0))
      (make-music
        'NoteEvent
        'articulations
        (list (make-music
                'SlurEvent
                'span-direction
                1))
        'duration
        (ly:make-duration 2 0 1/1)
        'pitch
        (ly:make-pitch 0 5 0))))

```

The bad news is that the `SlurEvent` expressions must be added ‘inside’ the note (in its `articulations` property).

Now we examine the input.

```

\displayMusic a'
====>
(make-music

```

```
'NoteEvent
'duration
(ly:make-duration 2 0 1/1)
'pitch
(ly:make-pitch 0 5 0)))))
```

So in our function, we need to clone this expression (so that we have two notes to build the sequence), add a `SlurEvent` to the `'articulations` property of each one, and finally make a `SequentialMusic` with the two `NoteEvent` elements. For adding to a property, it is useful to know that an unset property is read out as `'()`, the empty list, so no special checks are required before we put another element at the front of the `articulations` property.

```
doubleSlur = #(define-music-function (note) (ly:music?)
  "Return: { note ( note ) }.
  `note` is supposed to be a single note."
  (let ((note2 (ly:music-deep-copy note)))
    (set! (ly:music-property note 'articulations)
      (cons (make-music 'SlurEvent 'span-direction -1)
        (ly:music-property note 'articulations)))
    (set! (ly:music-property note2 'articulations)
      (cons (make-music 'SlurEvent 'span-direction 1)
        (ly:music-property note2 'articulations)))
    (make-music 'SequentialMusic 'elements (list note note2))))
```

1.3.5 Adding articulation to notes (example)

The easy way to add articulation to notes is to juxtapose two music expressions. However, suppose that we want to write a music function that does this.

A `$variable` inside the `#{...#}` notation is like a regular `\variable` in classical LilyPond notation. We could write

```
{ \music -. -> }
```

but for the sake of this example, we will learn how to do this in Scheme. We begin by examining our input and desired output.

```
% input
\displayMusic c4
====>
(make-music
  'NoteEvent
  'duration
  (ly:make-duration 2 0 1/1)
  'pitch
  (ly:make-pitch -1 0 0)))))
=====
% desired output
\displayMusic c4->
====>
(make-music
  'NoteEvent
```

```
'articulations
(list (make-music
      'ArticulationEvent
      'articulation-type 'accent))
'duration
(ly:make-duration 2 0 1/1)
'pitch
(ly:make-pitch -1 0 0))
```

We see that a note (c4) is represented as a NoteEvent expression. To add an accent articulation, an ArticulationEvent expression must be added to the articulations property of the NoteEvent expression.

To build this function, we begin with

```
(define (add-accent note-event)
  "Add an accent ArticulationEvent to the articulations of `note-event`,
  which is supposed to be a NoteEvent expression."
  (set! (ly:music-property note-event 'articulations)
        (cons (make-music 'ArticulationEvent
                          'articulation-type 'accent)
              (ly:music-property note-event 'articulations)))
  note-event)
```

The first line is the way to define a function in Scheme: the function name is add-accent, and has one variable called note-event. In Scheme, the type of variable is often clear from its name. (this is good practice in other programming languages, too!)

```
"Add an accent..."
```

is a description of what the function does. This is not strictly necessary, but just like clear variable names, it is good practice.

You may wonder why we modify the note event directly instead of working on a copy (ly:music-deep-copy can be used for that). The reason is a silent contract: music functions are allowed to modify their arguments: they are either generated from scratch (like user input) or are already copied (referencing a music variable with '\name' or music from immediate Scheme expressions '\$(...)' provides a copy). Since it would be inefficient to create unnecessary copies, the return value from a music function is *not* copied. So to heed that contract, you must not use any arguments more than once, and returning it counts as one use.

In an earlier example, we constructed music by repeating a given music argument. In that case, at least one repetition had to be a copy of its own. If it weren't, strange things may happen. For example, if you use \relative or \transpose on the resulting music containing the same elements multiple times, those will be subjected to relativation or transposition multiple times. If you assign them to a music variable, the curse is broken since referencing '\name' will again create a copy which does not retain the identity of the repeated elements.

Now while the above function is not a music function, it will normally be used within music functions. So it makes sense to heed the same contract we use for music functions: the input may be modified for producing the output, and the caller is responsible for creating copies if it still needs the unchanged argument itself. If you take a look at LilyPond's own functions like `music-map`, you'll find that they stick with the same principles.

Where were we? We now have a `note-event` we may modify, not because of using `ly:music-deep-copy` but because of a long-winded explanation. We add the accent to its `'articulations` list property.

```
(set! place new-value)
```

Here, what we want to set (the `'place`) is the `'articulations` property of `note-event` expression.

```
(ly:music-property note-event 'articulations)
```

`ly:music-property` is the function used to access music properties (the `'articulations`, `'duration`, `'pitch`, etc, that we see in the `\displayMusic` output above). The new value is the former `'articulations` property, with an extra item: the `ArticulationEvent` expression, which we copy from the `\displayMusic` output,

```
(cons (make-music 'ArticulationEvent
  'articulation-type 'accent)
  (ly:music-property result-event-chord 'articulations))
```

`cons` is used to add an element to the front of a list without modifying the original list. This is what we want: the same list as before, plus the new `ArticulationEvent` expression. The order inside the `'articulations` property is not important here.

Finally, once we have added the accent articulation to its `articulations` property, we can return `note-event`, hence the last line of the function.

Now we transform the `add-accent` function into a music function (a matter of some syntactic sugar and a declaration of the type of its argument).

```
addAccent = #(define-music-function (note-event) (ly:music?)
  "Add an accent ArticulationEvent to the articulations of `note-event`,
  which is supposed to be a NoteEvent expression."
  (set! (ly:music-property note-event 'articulations)
    (cons (make-music 'ArticulationEvent
      'articulation-type 'accent)
      (ly:music-property note-event 'articulations)))
  note-event)
```

We then verify that this music function works correctly:

```
\displayMusic \addAccent c4
```

Footnotes

(1) This artificial name was derived from the English word ‘breadth’, which means approximately the same as ‘width’.

2 Interfaces for programmers

Advanced tweaks may be performed by using Scheme. If you are not familiar with Scheme, you may wish to read our [Scheme tutorial](#).

2.1 LilyPond code blocks

Creating music expressions in Scheme can be tedious, as they are heavily nested and the resulting Scheme code is large. For some simple tasks this can be avoided by using LilyPond code blocks, which enable common LilyPond syntax to be used within Scheme.

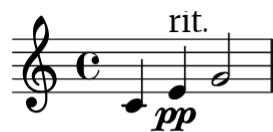
LilyPond code blocks look like

```
#{ LilyPond code #}
```

Here is a trivial example:

```
ritpp = #(define-event-function () ()
  #{ ^"rit." \pp #}
)

{ c'4 e'4\ritpp g'2 }
```



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LilyPond code blocks can be used anywhere where you can write Scheme code. The Scheme reader actually is changed for accommodating LilyPond code blocks and can deal with embedded Scheme expressions starting with '\$' and '#'.

The reader extracts the LilyPond code block and generates a runtime call to the LilyPond parser to interpret the LilyPond code. Scheme expressions embedded in the LilyPond code are evaluated in the lexical environment of the LilyPond code block, so all local variables and function parameters available at the point the LilyPond code block is written may be accessed. Variables defined in other Scheme modules, like the modules containing `\header` and `\layout` blocks, are not accessible as Scheme variables, i.e., prefixed with '#', but they are accessible as LilyPond variables, i.e., prefixed with '\'.

All music generated inside the code block has its 'origin' set to the current input location.

A LilyPond code block may contain anything that you can use on the right side of an assignment. In addition, an empty LilyPond block corresponds to a void music expression, and a LilyPond block containing multiple music events gets turned into a sequential music expression.

2.2 Scheme functions

Scheme functions are Scheme procedures that can create Scheme expressions from input written in LilyPond syntax. They can be called in pretty much all places where using ‘#’ for specifying a value in Scheme syntax is allowed. While Scheme has functions of its own, this chapter is concerned with *syntactic* functions, functions that receive arguments specified in LilyPond syntax.

2.2.1 Scheme function definitions

The general form for defining Scheme functions is:

```
function =
#(define-scheme-function
  (arg1 arg2 ...)
  (type1? type2? ...)
  body)
```

where

argN

nth argument.

typeN?

A Scheme *type predicate* for which *argN* must return `#t`. There is also a special form (*predicate? default*) for specifying optional arguments. If the actual argument is missing when the function is being called, the default value is substituted instead. Default values are evaluated at definition time (including LilyPond code blocks!), so if you need a default calculated at runtime, instead write a special value you can easily recognize. If you write the predicate in parentheses but don’t follow it with a default value, `#f` is used as the default. Default values are not verified with *predicate?* at either definition or run time: it is your responsibility to deal with the values you specify. Default values that happen to be music expressions are copied while setting `origin` to the current input location.

body

A sequence of Scheme forms evaluated in order, the last one being used as the return value of the Scheme function. It may contain LilyPond code blocks enclosed in hashed braces (`{...#}`), like described in [LilyPond code blocks](#). Within LilyPond code blocks, use ‘#’ to reference function arguments (e.g., `#arg1`) or to start an inline Scheme expression containing function arguments (e.g., `$(cons arg1 arg2)`). Where normal Scheme expressions using ‘#’ don’t do the trick, you might need to revert to immediate Scheme expressions using ‘\$’, for example as `$music`.

If your function returns a music expression, it is given a useful value of `origin`.

Suitability of arguments for the predicates is determined by actually calling the predicate after LilyPond has already converted them into a Scheme expression. As a consequence, the argument can be specified in Scheme syntax if desired (introduced with ‘#’ or as the result of calling a Scheme function), but LilyPond will also convert a number of LilyPond constructs into Scheme before actually checking the predicate on them. Currently, those include music, postevents, simple strings (with or without quotes), numbers, full markups and markup lists, score, book, bookpart, context definition and output definition blocks.

Some ambiguities LilyPond sorts out by checking with predicate functions: is ‘-3’ a fingering postevent or a negative number? Is “a” 4 in lyric mode a string followed by a number, or a lyric event of duration 4? LilyPond tries the argument predicate on successive interpretations until success, with an order designed to minimize inconsistent interpretations and lookahead.

For example, a predicate accepting both music expressions and pitches will consider `c` to be a pitch rather than a music expression. Immediately following durations or postevents will change that interpretation. It’s best to avoid overly permissive predicates like `scheme?` when the application rather calls for more specific argument types.

For a list of available predefined type predicates, see *Predefined type predicates in Notation Reference*.

See also

Notation Reference: *Predefined type predicates in Notation Reference*.

Installed Files: `lily/music-scheme.cc`, `scm/c++.scm`, `scm/lily.scm`.

2.2.2 Scheme function usage

Scheme functions can be called pretty much anywhere where a Scheme expression starting with ‘#’ can be written. You call a Scheme function from LilyPond by writing its name preceded by `\`, followed by its arguments. Once an optional argument predicate does not match an argument, LilyPond skips this and all following optional arguments, replacing them with their specified default, and ‘backs up’ the argument that did not match to the place of the next mandatory argument. Since the backed up argument needs to go somewhere, optional arguments are not actually considered optional unless followed by a mandatory argument.

There is one exception: if you write `\default` in the place of an optional argument, this and all following optional arguments are skipped and replaced by their default. This works even when no mandatory argument follows since `\default` does not need to get backed up. The `mark` and `key` commands make use of that trick to provide their default behavior when just followed by `\default`.

Apart from places where a Scheme value is required, there are a few places where ‘#’ expressions are currently accepted and evaluated for their side effects but otherwise ignored. Mostly those are the places where an assignment would be acceptable as well.

Since it is a bad idea to return values that can be misinterpreted in some context, you should use normal Scheme functions only for those cases where you always return a useful value, and use void Scheme functions (see [Void Scheme functions](#)) otherwise.

For convenience, Scheme functions may also be called directly from Scheme bypassing the LilyPond parser. Their name can be used like the name of an ordinary function. Typechecking of the arguments and skipping optional arguments will happen in the same manner as when called from within LilyPond, with the Scheme value `*unspecified*` taking the role of the `\default` reserved word for explicitly skipping optional arguments. Optional arguments at the end of an argument list can just be omitted without indication when called via Scheme.

2.2.3 Void Scheme functions

Sometimes a procedure is executed in order to perform an action rather than return a value. Some programming languages (like C and Scheme) use functions for either concept and just discard the returned value (usually by allowing any expression to act as statement, ignoring the result). This is clever but error-prone: most C compilers nowadays offer warnings for various non-‘void’ expressions being discarded. For many functions executing an action, the Scheme standards declare the return value to be unspecified. LilyPond’s Scheme interpreter Guile has a unique value `*unspecified*` that it usually (such when using `set!` directly on a variable) but unfortunately not consistently returns in such cases.

Defining a LilyPond function with `define-void-function` makes sure that this special value (the only value satisfying the predicate `void?`) will be returned.

```
noPointAndClick =
  #(define-void-function
    ()
    ()
    (ly:set-option 'point-and-click #f))
...
\noPointAndClick % disable point and click
```

If you want to evaluate an expression only for its side-effect and don’t want any value it may return interpreted, you can do so by prefixing it with `\void`:

```
\void #(hashq-set! some-table some-key some-value)
```

That way, you can be sure that LilyPond will not assign meaning to the returned value regardless of where it encounters it. This will also work for music functions such as `\displayMusic`.

2.3 Music functions

Music functions are Scheme procedures that can create music expressions automatically, and can be used to greatly simplify the input file.

2.3.1 Music function definitions

The general form for defining music functions is:

```
function =
#(define-music-function
  (arg1 arg2 ...)
  (type1? type2? ...)
  body)
```

quite in analogy to [Scheme function definitions](#). More often than not, *body* will be a [LilyPond code block](#).

For a list of available type predicates, see Predefined type predicates in *Notation Reference*.

See also

Notation Reference: Predefined type predicates in *Notation Reference*.

Installed Files: `lily/music-scheme.cc`, `scm/c++.scm`, `scm/lily.scm`.

2.3.2 Music function usage

With regard to handling the argument list, music functions don't differ from Scheme functions, [Scheme function usage](#).

A 'music function' has to return an expression matching the predicate `ly:music?`. This makes music function calls suitable as arguments of type `ly:music?` for another music function call.

When using a music function call in other contexts, the context may cause further semantic restrictions.

-

At the top level in a music expression a post-event is not accepted.

-

When a music function (as opposed to an event function) returns an expression of type post-event, LilyPond requires one of the named direction indicators (`-`, `^`, and `_`) in order to properly integrate the post-event produced by the music function call into the surrounding expression.

-

As a chord constituent. The returned expression must be of a `rhythmic-event` type, most likely a `NoteEvent`.

'Polymorphic' functions, like `\tweak`, can be applied to post-events, chord constituent and top level music expressions.

2.3.3 Simple substitution functions

Simple substitution functions are music functions whose output music expression is written in LilyPond format and contains function arguments in the output expression. They are described in Substitution function examples in *Notation Reference*.

2.3.4 Intermediate substitution functions

Intermediate substitution functions involve a mix of Scheme code and LilyPond code in the music expression to be returned.

Some `\override` commands require an argument consisting of a pair of numbers (called a *cons cell* in Scheme).

The pair can be directly passed into the music function, using a `pair?` variable:

```
manualBeam =
#(define-music-function
  (beg-end)
  (pair?)
  #{
    \once \override Beam.positions = #beg-end
  })

\relative c' {
  \manualBeam #'(3 . 6) c8 d e f
}
```

Alternatively, the numbers making up the pair can be passed as separate arguments, and the Scheme code used to create the pair can be included in the music expression:

```
manualBeam =
#(define-music-function
  (beg end)
  (number? number?)
  #{
    \once \override Beam.positions = #(cons beg end)
  })

\relative c' {
  \manualBeam #3 #6 c8 d e f
}
```



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Properties are maintained conceptually using one stack per property per grob per context. Music functions may need to override one or several properties for the duration of the function, restoring them to their previous value before exiting. However, normal

overrides pop and discard the top of the current property stack before pushing to it, so the previous value of the property is lost when it is overridden. When the previous value must be preserved, prefix the `\override` command with `\temporary`, like this:

```
\temporary \override ...
```

The use of `\temporary` causes the (usually set) pop-first property in the override to be cleared, so the previous value is not popped off the property stack before pushing the new value onto it. When a subsequent `\revert` pops off the temporarily overridden value, the previous value will re-emerge.

In other words, calling `\temporary \override` and `\revert` in succession on the same property will have a net effect of zero. Similarly, pairing `\temporary` and `\undo` on the same music containing overrides will have a net effect of zero.

Here is an example of a music function which makes use of this. The use of `\temporary` ensures the values of the `cross-staff` and `style` properties are restored on exit to whatever values they had when the `crossStaff` function was called. Without `\temporary` the default values would have been set on exit.

```
crossStaff =
#(define-music-function (notes) (ly:music?)
  (_i "Create cross-staff stems")
  #{
    \temporary \override Stem.cross-staff = #cross-staff-connect
    \temporary \override Flag.style = #'no-flag
    #notes
    \revert Stem.cross-staff
    \revert Flag.style
  #})
```

2.3.5 Mathematics in functions

Music functions can involve Scheme programming in addition to simple substitution,

```
AltOn =
#(define-music-function
  (mag)
  (number?)
  #{
    \override Stem.length =>(* 7.0 mag)
    \override NoteHead.font-size =
      #(inexact->exact (* (/ 6.0 (log 2.0)) (log mag)))
  #})

AltOff = {
  \revert Stem.length
  \revert NoteHead.font-size
}

\relative {
  c'2 \AltOn #0.5 c4 c
  \AltOn #1.5 c c \AltOff c2
```

}



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This example may be rewritten to pass in music expressions,

```
withAlt =
#(define-music-function
  (mag music)
  (number? ly:music?)
  #{
    \override Stem.length = #(* 7.0 mag)
    \override NoteHead.font-size =
      #(inexact->exact (* (/ 6.0 (log 2.0)) (log mag)))
    #music
    \revert Stem.length
    \revert NoteHead.font-size
  #})

\relative {
  c'2 \withAlt #0.5 { c4 c }
  \withAlt #1.5 { c c } c2
}
```



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2.3.6 Functions without arguments

In most cases a function without arguments should be written with a variable,

```
dolce = \markup{ \italic \bold dolce }
```

However, in rare cases it may be useful to create a music function without arguments,

```
displayBarNum =
#(define-music-function
  ()
  ()
  (if (eq? #t (ly:get-option 'display-bar-numbers))
    #{ \once \override Score.BarNumber.break-visibility = ##f #}
    #{#}))
```

To actually display bar numbers where this function is called, invoke lilypond with

```
lilypond -d display-bar-numbers FILENAME.ly
```

2.3.7 Void music functions

A music function must return a music expression. If you want to execute a function only for its side effect, you should use `define-void-function`. But there may be cases where you sometimes want to produce a music expression, and sometimes not (like in the previous example). Returning a void music expression via `#{ # }` will achieve that.

2.4 Event functions

To use a music function in the place of an event, you need to write a direction indicator before it. But sometimes, this does not quite match the syntax of constructs you want to replace. For example, if you want to write dynamics commands, those are usually attached without direction indicator, like `c' \pp`. Here is a way to write arbitrary dynamics:

```
dyn=#(define-event-function (arg) (markup?)
      (make-dynamic-script arg))
\relative { c' \dyn pfsss }
```



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You could do the same using a music function, but then you always would have to write a direction indicator before calling it, like `c- \dyn pfsss`.

2.5 Markup functions

Markups are implemented as special Scheme functions which produce a `Stencil` object given a number of arguments.

2.5.1 Markup construction in Scheme

Markup expressions are internally represented in Scheme using the `markup` macro:

```
(markup expr)
```

To see a markup expression in its Scheme form, use the `\displayScheme` command:

```
\displayScheme
\markup {
  \column {
    \line { \bold \italic "hello" \raise #0.4 "world" }
    \larger \line { foo bar baz }
  }
}
```

Compiling the code above will send the following to the display console:

```
(markup
  #:line
  (#:column
    (#:line
      (#:bold (#:italic "hello") #:raise 0.4 "world")
      #:larger
      (#:line ("foo" "bar" "baz")))))
```

To prevent the markup from printing on the page, use ‘`\void \displayScheme markup`’. Also, as with the `\displayMusic` command, the output of `\displayScheme` can be saved to an external file. See [Displaying music expressions](#).

This example demonstrates the main translation rules between regular LilyPond markup syntax and Scheme markup syntax. Using `#{ ... #}` for entering in LilyPond syntax will often be most convenient, but we explain how to use the `markup` macro to get a Scheme-only solution.

LilyPond	Scheme
<code>\markup markup1</code>	<code>(markup markup1)</code>
<code>\markup { markup1 markup2 ... }</code>	<code>(markup markup1 markup2 ...)</code>
<code>\markup-command</code>	<code>#:markup-command</code>
<code>\variable</code>	<code>variable</code>
<code>\center-column { ... }</code>	<code>#:center-column (...)</code>
<code>string</code>	<code>"string"</code>
<code>#scheme-arg</code>	<code>scheme-arg</code>

The whole Scheme language is accessible inside the `markup` macro. For example, You may use function calls inside `markup` in order to manipulate character strings. This is useful when defining new markup commands (see [New markup command definition](#)).

Known issues and warnings

The `markup`-list argument of commands such as `#:line`, `#:center-column`, and `#:column` cannot be a variable or the result of a function call.

```
(markup #:line (function-that-returns-markups))
```

is invalid. One should use the `make-line-markup`, `make-center-column-markup`, or `make-column-markup` functions instead,

```
(markup (make-line-markup (function-that-returns-markups)))
```

2.5.2 How markups work internally

In a markup like

```
\raise #0.5 "text example"
```

`\raise` is actually represented by the `raise-markup` function. The markup expression is stored as

```
(list raise-markup 0.5 "text example")
```

When the markup is converted to printable objects (Stencils), the `raise-markup` function is called as

```
(apply raise-markup
      \layout object
      list of property alists
      0.5
      the "text example" markup)
```

The `raise-markup` function first creates the stencil for the `text example` string, and then it raises that Stencil by 0.5 staff space. This is a rather simple example; more complex examples are in the rest of this section, and in `scm/define-markup-commands.scm`.

2.5.3 New markup command definition

This section discusses the definition of new markup commands.

Markup command definition syntax

New markup commands can be defined using the `define-markup-command` Scheme macro, at top-level.

```
(define-markup-command (command-name layout props arg1 arg2 ...)
  (arg1-type? arg2-type? ...)
  [ #:properties ((property1 default-value1)
                  ...) ]
  [ #:as-string expression ]
  ...command body...)
```

The arguments are

command-name

the markup command name

layout

the ‘layout’ definition.

props

a list of associative lists, containing all active properties.

argi

*i*th command argument

argi-type?

a type predicate for the *i*th argument

If the command uses properties from the `props` arguments, the `#:properties` keyword can be used to specify which properties are used along with their default values.

Arguments are distinguished according to their type:

- a markup, corresponding to type predicate `markup?`;
- a list of markups, corresponding to type predicate `markup-list?`;
- any other Scheme object, corresponding to type predicates such as `list?`, `number?`, `boolean?`, etc.

There is no limitation on the order of arguments (after the standard `layout` and `props` arguments). However, markup functions taking a markup as their last argument are somewhat special as you can apply them to a markup list, and the result is a markup list where the markup function (with the specified leading arguments) has been applied to every element of the original markup list.

Since replicating the leading arguments for applying a markup function to a markup list is cheap mostly for Scheme arguments, you avoid performance pitfalls by just using Scheme arguments for the leading arguments of markup functions that take a markup as their last argument.

Markup commands have a rather complex life cycle. The body of a markup command definition is responsible for converting the arguments of the markup command into a stencil expression which is returned. Quite often this is accomplished by calling the `interpret-markup` function on a markup expression, passing the *layout* and *props* arguments on to it. Those arguments are usually only known at a very late stage in typesetting. Markup expressions have their components assembled into markup expressions already when `\markup` in a LilyPond expression or the `markup` macro in Scheme is expanded. The evaluation and typechecking of markup command arguments happens at the time `\markup/markup` are interpreted.

But the actual conversion of markup expressions into stencil expressions by executing the markup function bodies only happens when `interpret-markup` is called on a markup expression.

On properties

The `layout` and `props` arguments of markup commands bring a context for the markup interpretation: font size, line width, etc.

The `layout` argument allows access to properties defined in paper blocks, using the `ly:output-def-lookup` function. For instance, the line width (the same as the one used in scores) is read using:

```
(ly:output-def-lookup layout 'line-width)
```

The `props` argument makes some properties accessible to markup commands. For instance, when a book title markup is interpreted, all the variables defined in the `\header` block are automatically added to `props`, so that the book title markup can access the book title, composer, etc. It is also a way to configure the behavior of a markup command: for example, when a command uses font size during processing, the font size is read from `props` rather than having a `font-size` argument. The caller of a markup command may change the value of the font size property in order to change the behavior. Use the `#:properties` keyword of `define-markup-command` to specify which properties shall be read from the `props` arguments.

The example in next section illustrates how to access and override properties in a markup command.

A complete example

The following example defines a markup command to draw a double box around a piece of text.

Firstly, we need to build an approximative result using markups. Consulting the Text markup commands in *Notation Reference* shows us the `\box` command is useful:

```
\markup \box \box HELLO
```

HELLO

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Now, we consider that more padding between the text and the boxes is preferable. According to the `\box` documentation, this command uses a `box-padding` property, which defaults to 0.2. The documentation also mentions how to override it:

```
\markup \box \override #'(box-padding . 0.6) \box A
```

A

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Then, the padding between the two boxes is considered too small, so we override it too:

```
\markup \override #'(box-padding . 0.4) \box  
  \override #'(box-padding . 0.6) \box A
```

A

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Repeating this lengthy markup would be painful. This is where a markup command is needed. Thus, we write a double-box markup command, taking one argument (the text). This draws the two boxes, with some padding.

```
#(define-markup-command (double-box layout props text) (markup?)
  "Draw a double box around text."
  (interpret-markup layout props
    #{\markup \override #'(box-padding . 0.4) \box
      \override #'(box-padding . 0.6) \box { #text }#}))
```

or, equivalently

```
#(define-markup-command (double-box layout props text) (markup?)
  "Draw a double box around text."
  (interpret-markup layout props
    (markup #:override '(box-padding . 0.4) #:box
      #:override '(box-padding . 0.6) #:box text)))
```

text is the name of the command argument, and markup? its type: it identifies it as a markup. The interpret-markup function is used in most of markup commands: it builds a stencil, using layout, props, and a markup. In the second case, this markup is built using the markup Scheme macro, see [Markup construction in Scheme](#). The transformation from \markup expression to Scheme markup expression is straight-forward.

The new command can be used as follow:

```
\markup \double-box A
```

It would be nice to make the double-box command customizable: here, the box-padding values are hard coded, and cannot be changed by the user. Also, it would be better to distinguish the padding between the two boxes, from the padding between the inner box and the text. So we will introduce a new property, inter-box-padding, for the padding between the two boxes. The box-padding will be used for the inner padding. The new code is now as follows:

```
#(define-markup-command (double-box layout props text) (markup?)
  #:properties ((inter-box-padding 0.4)
    (box-padding 0.6))
  "Draw a double box around text."
  (interpret-markup layout props
    #{\markup \override #`(box-padding . ,inter-box-padding) \box
      \override #`(box-padding . ,box-padding) \box
        { #text } #}))
```

Again, the equivalent version using the markup macro would be:

```
#(define-markup-command (double-box layout props text) (markup?)
  #:properties ((inter-box-padding 0.4)
    (box-padding 0.6))
  "Draw a double box around text."
  (interpret-markup layout props
```



```
(markup #:override `(box-padding . ,inter-box-padding) #:box
      #:override `(box-padding . ,box-padding) #:box text)))
```

Here, the `#:properties` keyword is used so that the `inter-box-padding` and `box-padding` properties are read from the `props` argument, and default values are given to them if the properties are not defined.

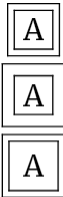
Then, these values are used to override the `box-padding` properties used by the two `\box` commands. Note the backquote and the comma in the `\override` argument: they allow you to introduce a variable value into a literal expression.

Now, the command can be used in a markup, and the boxes padding be customized:

```

(define-markup-command (double-box layout props text) (markup?)
  #:properties ((inter-box-padding 0.4)
                (box-padding 0.6))
  "Draw a double box around text."
  (interpret-markup layout props
    #{\markup \override #`(box-padding . ,inter-box-padding) \box
      \override #`(box-padding . ,box-padding) \box
      { #text } #}))

\markup \double-box A
\markup \override #'(inter-box-padding . 0.8) \double-box A
\markup \override #'(box-padding . 1.0) \double-box A
    
```



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Adapting builtin commands

A good way to start writing a new markup command, is to take example on a builtin one. Most of the markup commands provided with LilyPond can be found in file `scm/define-markup-commands.scm`.

For instance, we would like to adapt the `\draw-line` command, to draw a double line instead. The `\draw-line` command is defined as follow (documentation stripped):

```

(define-markup-command (draw-line layout props dest)
  (number-pair?)
  #:category graphic
  #:properties ((thickness 1))
  "...documentation..."
  (let ((th (* (ly:output-def-lookup layout 'line-thickness)
               thickness))
        (x (car dest))
        (y (cdr dest)))
    
```

```
(make-line-stencil th 0 0 x y)))
```

To define a new command based on an existing one, copy the definition, and change the command name. The `#:category` keyword can be safely removed, as it is only used for generating LilyPond documentation, and is of no use for user-defined markup commands.

```
(define-markup-command (draw-double-line layout props dest)
  (number-pair?)
  #:properties ((thickness 1)
    "...documentation...")
  (let ((th (* (ly:output-def-lookup layout 'line-thickness)
    thickness))
    (x (car dest))
    (y (cdr dest)))
    (make-line-stencil th 0 0 x y)))
```

Then, a property for setting the gap between two lines is added, called `line-gap`, defaulting, e.g., to 0.6:

```
(define-markup-command (draw-double-line layout props dest)
  (number-pair?)
  #:properties ((thickness 1)
    (line-gap 0.6))
  "...documentation..."
  ...)
```

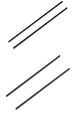
Finally, the code for drawing two lines is added. Two calls to `make-line-stencil` are used to draw the lines, and the resulting stencils are combined using `ly:stencil-add`:

```

#(define-markup-command (my-draw-line layout props dest)
  (number-pair?)
  #:properties ((thickness 1)
    (line-gap 0.6))
  "...documentation..."
  (let* ((th (* (ly:output-def-lookup layout 'line-thickness)
    thickness))
    (dx (car dest))
    (dy (cdr dest))
    (w (/ line-gap 2.0))
    (x (cond ((= dx 0) w)
      ((= dy 0) 0)
      (else (/ w (sqrt (+ 1 (* (/ dx dy) (/ dx dy))))))))
    (y (cond ((= dy 0) w)
      ((= dx 0) 0)
      (else (/ w (sqrt (+ 1 (* (/ dy dx) (/ dy dx))))))))
    (y (* (if (< (* dx dy) 0) 1 -1) y)))
    (ly:stencil-add
      (make-line-stencil th x y (+ dx x) (+ dy y))
      (make-line-stencil th (- x) (- y) (- dx x) (- dy y))))

\markup \my-draw-line #'(4 . 3)
\markup \override #'(line-gap . 1.2) \my-draw-line #'(4 . 3)

```



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Converting markups to strings

Markups are occasionally converted to plain strings, such as when outputting PDF metadata based on the `title` header field or for converting lyrics to MIDI. This conversion is inherently lossy, but tries to as accurate as feasible. The function used for this is `markup->string`.

```
composerName = \markup \box "Arnold Schönberg"

\markup \composerName

\markup \typewriter #(markup->string composerName)
```

Arnold Schönberg
[Arnold Schönberg]

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For custom markup commands, the default behavior is to convert all markup or markup list arguments first, and join the results by spaces.

```
#{(define-markup-command (authors-and layout props authorA authorB)
  (markup? markup?)
  (interpret-markup layout props
    #{
      \markup \fill-line { \box #authorA and \box #authorB }
    #}))

defaultBehavior = \markup \authors-and "Bertolt Brecht" "Kurt Weill"

\markup \defaultBehavior

\markup \typewriter #(markup->string defaultBehavior)
```

Bertolt Brecht

and

Kurt Weill

Bertolt Brecht Kurt Weill

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`markup->string` can also receive the named arguments `#:layout` *layout* and `#:props` *props*, with the same meaning as in the definition of a markup command. However, they are optional because they cannot always be provided (such as the `layout` argument when converting to MIDI).

To support special conversions in custom markup commands, the `#:as-string` parameter

can be given to define-markup-command. It expects an expression, which is evaluated by markup->string in order to yield the string result.

```
#(define-markup-command (authors-and layout props authorA authorB)
  (markup? markup?)
  #:as-string (format
    #f "~a and ~a"
    (markup->string authorA #:layout layout #:props props)
    (markup->string authorB #:layout layout #:props props))
  (interpret-markup layout props
    #{
      \markup \fill-line { \box #authorA and \box #authorB }
    })))

customized = \markup \authors-and "Bertolt Brecht" "Kurt Weill"

\markup \customized

\markup \typewriter #(markup->string customized)
```

Bertolt Brecht and Kurt Weill
Bertolt Brecht and Kurt Weill

LilyPond v2.27.2

Within the expression, the same bindings are available as in the main markup command body, namely layout and props, the command argument, and optionally the properties.

```
#(define-markup-command (authors-and layout props authorA authorB)
  (markup? markup?)
  #:properties ((author-separator " and "))
  #:as-string (format
    #f "~a~a~a"
    (markup->string authorA
      #:layout layout #:props props)
    (markup->string author-separator
      #:layout layout #:props props)
    (markup->string authorB
      #:layout layout #:props props))
  (interpret-markup layout props
    #{
      \markup { \box #authorA #author-separator \box #authorB }
    })))

customized = \markup \override #'(author-separator . ", ")
  \authors-and "Bertolt Brecht" "Kurt Weill"

\markup \customized

\markup \typewriter #(markup->string customized)
```

Bertolt Brecht, Kurt Weill
Bertolt Brecht, Kurt Weill

LilyPond v2.27.2

Most often, a custom handler only needs to call `markup->string` recursively on certain arguments, as demonstrated above. However, it can also make use of the `layout` and `props` directly, in the same way as in the main body. Care must be taken that the `layout` argument is `#f` if no `#:layout` has been given to `markup->string`. The `props` argument defaults to the empty list.

2.5.4 New markup list command definition

Markup list commands are defined with the `define-markup-list-command` Scheme macro, which is similar to the `define-markup-command` macro described in [New markup command definition](#), except that where the latter returns a single stencil, the former returns a list of stencils.

In a similar vein, `interpret-markup-list` is used instead of `interpret-markup` for converting a markup list into a list of stencils.

In the following example, a `\paragraph` markup list command is defined, which returns a list of justified lines, the first one being indented. The indent width is taken from the `props` argument.

```
#(define-markup-list-command (paragraph layout props args) (markup-list?)
  #:properties ((par-indent 2))
  (interpret-markup-list layout props
    #{\markuplist \justified-lines { \hspace #par-indent #args } #}))
```

The version using just Scheme is more complex:

```
#(define-markup-list-command (paragraph layout props args) (markup-list?)
  #:properties ((par-indent 2))
  (interpret-markup-list layout props
    (make-justified-lines-markup-list (cons (make-hspace-markup par-indent)
      args))))
```

Besides the usual `layout` and `props` arguments, the `paragraph` markup list command takes a markup list argument, named `args`. The predicate for markup lists is `markup-list?`.

First, the function gets the indent width, a property here named `par-indent`, from the property list `props`. If the property is not found, the default value is 2. Then, a list of justified lines is made using the built-in markup list command `\justified-lines`, which is related to the `make-justified-lines-markup-list` function. A horizontal space is added at the beginning using `\hspace` (or the `make-hspace-markup` function). Finally, the markup list is interpreted using the `interpret-markup-list` function.

This new markup list command can be used as follows:

```
\markuplist {
  \paragraph {
    The art of music typography is called \italic {(plate) engraving.}
    The term derives from the traditional process of music printing.
    Just a few decades ago, sheet music was made by cutting and stamping
    the music into a zinc or pewter plate in mirror image.
```

```

}
\override-lines #'(par-indent . 4) \paragraph {
  The plate would be inked, the depressions caused by the cutting
  and stamping would hold ink. An image was formed by pressing paper
  to the plate. The stamping and cutting was completely done by
  hand.
}
}

```

2.6 Contexts for programmers

2.6.1 Context evaluation

Contexts can be modified during interpretation with Scheme code. In a LilyPond code block, the syntax for this is:

```
\applyContext function
```

In Scheme code, the syntax is:

```
(make-apply-context function)
```

function should be a Scheme function that takes a single argument: the context in which the `\applyContext` command is being called. The function can access as well as override/set grob properties and context properties. Any actions taken by the function that depend on the state of the context are limited to the state of the context *when the function is called*. Also, changes effected by a call to `\applyContext` remain in effect until they are directly modified again, or reverted, even if the initial conditions that they depended on have changed.

The following Scheme functions are useful when using `\applyContext`:

ly:context-property

look up a context property value

ly:context-set-property!

set a context property

ly:context-grob-definition

ly:assoc-get

look up a grob property value

ly:context-pushpop-property

do a `\temporary` `\override` or a `\revert` on a grob property

The following example looks up the current `fontSize` value, and then doubles it:

```

doubleFontSize =
\applyContext
  #(lambda (context)
    (let ((fontSize (ly:context-property context 'fontSize)))

```

```
(ly:context-set-property! context 'fontSize (+ fontSize 6)))

{
  \set fontSize = -3
  b'4
  \doubleFontSize
  b'
}
```



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The following example looks up the current colors of the NoteHead, Stem, and Beam grobs, and then changes each to a less saturated shade.

```
desaturate =
\applyContext
  #(lambda (context)
    (define (desaturate-grob grob)
      (let* ((grob-def (ly:context-grob-definition context grob))
        (color (ly:assoc-get 'color grob-def black))
        (new-color (map (lambda (x)
          (min 1 (/ (1+ x) 2)))
            color)))
        (ly:context-pushpop-property context grob
          'color new-color)))
      (for-each desaturate-grob '(NoteHead Stem Beam)))

\relative {
  \time 3/4
  g'8[ g] \desaturate g[ g] \desaturate g[ g]
  \override NoteHead.color = #darkred
  \override Stem.color = #darkred
  \override Beam.color = #darkred
  g[ g] \desaturate g[ g] \desaturate g[ g]
}
```



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This also could be implemented as a music function, in order to restrict the modifications to a single music block. Notice how `ly:context-pushpop-property` is used both as a `\temporary \override` and as a `\revert`:

```
desaturate =
#(define-music-function (music) (ly:music?)
  #{
    \applyContext
    #(lambda (context)
      (define (desaturate-grob grob)
```

```

    (let* ((grob-def (ly:context-grob-definition context grob))
           (color (ly:assoc-get 'color grob-def black))
           (new-color (map (lambda (x)
                             (min 1 (/ (1+ x) 2)))
                           color)))
           (ly:context-pushpop-property context grob
                                         'color new-color)))
    (for-each desaturate-grob '(NoteHead Stem Beam)))
#music
\applyContext
#(lambda (context)
    (define (revert-color grob)
      (ly:context-pushpop-property context grob 'color))
    (for-each revert-color '(NoteHead Stem Beam)))
#})

\relative {
  \override NoteHead.color = #darkblue
  \override Stem.color = #darkblue
  \override Beam.color = #darkblue
  g'8 a b c
  \desaturate { d c b a }
  g b d b g2
}

```



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2.6.2 Running a function on all layout objects

The most versatile way of tuning an object is `\applyOutput` which works by inserting an event into the specified context (`ApplyOutputEvent` in *Internals Reference*). Its syntax is either

```
\applyOutput Context proc
```

or

```
\applyOutput Context.Grob proc
```

where *proc* is a Scheme function, taking three arguments.

When interpreted, the function *proc* is called for every layout object (with grob name *Grob* if specified) found in the context *Context* at the current time step, with the following arguments:

-
- the layout object itself,
-

the context where the layout object was created, and

the context where `\applyOutput` is processed.

In addition, the cause of the layout object, i.e., the music expression or object that was responsible for creating it, is in the object property `cause`. For example, for a note head, this is a `NoteHead` in *Internals Reference* event, and for a stem object, this is a `Stem` in *Internals Reference* object.

Here is a function to use for `\applyOutput`; it blanks note-heads on the center-line and next to it:

```
#(define (blanker grob grob-origin context)
  (if (< (abs (ly:grob-property grob 'staff-position)) 2)
      (set! (ly:grob-property grob 'transparent) #t)))

\relative {
  a'4 e8 <<\applyOutput Voice.NoteHead #blanker a c d>> b2
}
```



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To have *function* interpreted at the `Score` or `Staff` level use these forms

```
\applyOutput Score...
\applyOutput Staff...
```

2.7 Callback functions

Properties (like `thickness`, `direction`, etc.) can be set at fixed values with `\override`, e.g.

```
\override Stem.thickness = #2.0
```

Properties can also be set to a Scheme procedure:

```
\override Stem.thickness = #(lambda (grob)
  (if (= UP (ly:grob-property grob 'direction))
      2.0
      7.0))
\relative { c'' b a g b a g b }
```



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In this case, the procedure is executed as soon as the value of the property is requested during the formatting process.

Most of the typesetting engine is driven by such callbacks. Properties that typically use callbacks include

stencil

The printing routine, that constructs a drawing for the symbol

X-offset

The routine that sets the horizontal position

X-extent

The routine that computes the width of an object

The procedure always takes a single argument, being the grob.

That procedure may access the usual value of the property, by first calling the function that is the usual callback for that property, which can be found in the Internals Reference or the file 'define-grobs.scm':

```
\relative {
  \override Flag.X-offset = #(lambda (flag)
    (let ((default (ly:flag::calc-x-offset flag)))
      (* default 4.0)))
  c' '4. d8 a4. g8
}
```

It is also possible to get the value of the existing default by employing the function `grob-transformer`:

```
\relative {
  \override Flag.X-offset = #(grob-transformer 'X-offset
    (lambda (flag default) (* default 4.0)))
  c' '4. d8 a4. g8
}
```



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From within a callback, the easiest method for evaluating a markup is to use `grob-interpret-markup`. For example:

```
my-callback = #(lambda (grob)
  (grob-interpret-markup grob (markup "foo")))
```

2.8 Unpure-pure containers

Note: While this subsection reflects the current state, the given example is moot and does not show any effect.

Unpure-pure containers are useful for overriding *Y-axis* spacing calculations – specifically *Y-offset* and *Y-extent* – with a Scheme function instead of a literal (i.e., a number or pair).

For certain grobs, the *Y-extent* is based on the `stencil` property, overriding the `stencil` property of one of these will require an additional *Y-extent* override with an unpure-pure container. When a function overrides a *Y-offset* and/or *Y-extent* it is assumed that this will trigger line breaking calculations too early during compilation. So the function is not evaluated at all (usually returning a value of `'0`' or `'(0 . 0)'`) which can result in collisions. A 'pure' function will not affect properties, objects or grob suicides and therefore will always have its *Y-axis*-related evaluated correctly.

Currently, there are about thirty functions that are already considered 'pure' and Unpure-pure containers are a way to set functions not on this list as 'pure'. The 'pure' function is evaluated *before* any line-breaking and so the horizontal spacing can be adjusted 'in time'. The 'unpure' function is then evaluated *after* line breaking.

Note: As it is difficult to always know which functions are on this list we recommend that any 'pure' functions you create do not use `Beam` or `VerticalAlignment` grobs.

An unpure-pure container is constructed as follows;

```
(ly:make-unpure-pure-container f0 f1)
```

where `f0` is a function taking n arguments ($n \geq 1$) and the first argument must always be the grob. This is the function that gives the actual result. `f1` is the function being labeled as 'pure' that takes $n + 2$ arguments. Again, the first argument must always still be the grob but the second and third are 'start' and 'end' arguments.

start and *end* are, for all intents and purposes, dummy values that only matter for *Spanners* (i.e. *Hairpin* or *Beam*), that can return different height estimations based on a starting and ending column.

The rest are the other arguments to the first function (which may be none if $n = 1$).

The results of the second function are used as an approximation of the value needed which is then used by the first function to get the real value which is then used for fine-tuning much later during the spacing process.

```
#(define (square-line-circle-space grob)
  (let* ((pitch (ly:event-property (ly:grob-property grob 'cause)
                                   'pitch))
        (notename (ly:pitch-notename pitch)))
    (if (= 0 (modulo notename 2))
```

```

(make-circle-stencil 0.5 0.0 #t)
(make-filled-box-stencil '(0 . 1.0)
                          '(-0.5 . 0.5))))

squareLineCircleSpace = {
  \override NoteHead.stencil = #square-line-circle-space
}

smartSquareLineCircleSpace = {
  \squareLineCircleSpace
  \override NoteHead.Y-extent =
    #(ly:make-unpure-pure-container
      ly:grob::stencil-height
      (lambda (grob start end) (ly:grob::stencil-height grob)))
}

\new Voice \with { \remove Stem_engraver }
\relative c'' {
  \squareLineCircleSpace
  cis4 ces disis d
  \smartSquareLineCircleSpace
  cis4 ces disis d
}

```



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In the first measure, without the unpure–pure container, the spacing engine does not know the width of the note head and lets it collide with the accidentals. In the second measure, with unpure–pure containers, the spacing engine knows the width of the note heads and avoids the collision by lengthening the line accordingly.

Usually for simple calculations nearly-identical functions for both the ‘unpure’ and ‘pure’ parts can be used, by only changing the number of arguments passed to, and the scope of, the function. This use case is frequent enough that `ly:make-unpure-pure-container` constructs such a second function by default when called with only one function argument.

Note: If a function is labeled as ‘pure’ and it turns out not to be, the results can be unexpected.

2.9 Difficult tweaks

There are a few classes of difficult adjustments.

.

One type of difficult adjustment involves the appearance of spanner objects, such as slurs and ties. Usually, only one spanner object is created at a time, and it can be adjusted with the normal mechanism. However, occasionally a spanner crosses a line

break. When this happens, the object is cloned. A separate object is created for every system in which the spanner appears. The new objects are clones of the original object and inherit all properties, including `\overrides`.

In other words, an `\override` always affects all pieces of a broken spanner. To change only one part of a spanner at a line break, it is necessary to hook into the formatting process. The `after-line-breaking` callback contains the Scheme procedure that is called after the line breaks have been determined and layout objects have been split over different systems.

In the following example, we define a procedure `my-callback`. This procedure

- determines if the spanner has been split across line breaks
- if yes, retrieves all the split objects
- checks if this grob is the last of the split objects
- if yes, it sets `extra-offset`.

This procedure is installed into `Tie` in *Internals Reference*, so the last part of the broken tie is repositioned.

```
#(define (my-callback grob)
  (let* (
    ;; have we been split?
    (orig (ly:grob-original grob))

    ;; if yes, get the split pieces (our siblings)
    (siblings (if (ly:grob? orig)
                  (ly:spanner-broken-into orig)
                  '())))

    (if (and (>= (length siblings) 2)
          (eq? (car (last-pair siblings)) grob))
        (ly:grob-set-property! grob 'extra-offset '(1 . -4))))

  \relative {
    \override Tie.after-line-breaking =
    #my-callback
    c'1 ~ \break
    c2 ~ 2
  }
```



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When applying this trick, the new after-line-breaking callback should also call the old one, if such a default exists. For example, if using this with `Hairpin`, `ly:spanner::kill-zero-spanned-time` should also be called.

•

Some objects cannot be changed with `\override` for technical reasons. They can be changed with the `\overrideProperty` function, which works similar to `\once \override`, but uses a different syntax.²

```
\overrideProperty [ContextName]
                  .GrobName
                  .property-name
                  [.subproperty-name] value
```

Footnotes

(2) Over time, `\overrideProperty` is being replaced with the more usual `\once \override` command. If you still find yourself needing `\overrideProperty`, a bug report would be appreciated.

3 LilyPond Scheme interfaces

This chapter covers the various tools provided by LilyPond to help Scheme programmers get information into and out of the music streams.

TODO – figure out what goes in here and how to organize it

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